ME2 - Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge, no cloud round-trip

Leandro Soneja - AI 231 ME2 | configuration: **gate1v12_f16 x gate2v20_aux - raw - architecture v1**

Model

Gate 1 (wake)	gate1v12_f16 - 513,636 B (0.49 MB)
Gate 2 (intent)	gate2v20_aux
Gate 2 params	246,035 intent-only (248,700 with the training-only aux head)
Gate 2 weights	515,332 B (0.52 MB, float16)
Noise canceller	raw (off)
Architecture	v1
Runtime	tflite-runtime, XNNPACK delegate

V19 TC-ResNet intent path + phoneme-auxiliary CTC head removed at inference. Clean synthetic val 98.84% (3,201 clips).

Dataset

Primary	class gold /data/ai231 + HF supplementals
Labels	20 intents / slots (frozen labels.json)
Wake words	Nova, Piattos (project-own schema)
Frozen split	SPLIT_LOCK.json, seed 42 - never modified
Takes scored	425 takes in this run

Measured results - this configuration

Full pipeline (wake + intent)	76.2% (157/206)
Intent accuracy (after wake)	93.6% (367/392)
Wake detection	79.1% (163/206)
Out-of-scope rejection	50.0% (16/32) (32 OOS takes)
Latency p95 / mean (host CPU)	72.4 ms / 51.3 ms
Raspberry Pi 4/5 latency	not measured for this lane (host CPU only)

Control: the same takes scored with the published reference Gate 1

gate1v3_f32 x gate2v20_aux - raw - v1	full 90.3% (186/206) intent 93.6% (367/392) p95 39.3 ms

Both rows were produced by one controlled pass over the identical take set, so the comparison is like-for-like. Latency is host-CPU p95 across takes; it is not a Raspberry Pi measurement.

Reproduction

Repository: https://github.com/lcsoneja/me2-voice-controlled-smart-device
Dashboard: `python gate1_server.py` then open `http://127.0.0.1:8765/`
This run: `py -3.12 _batch_v12_compare.py` then `py -3.12 _v12_analyze.py`

To be submitted - what the reviewer will look for

#	Item	Status
1	Repository public, one-command reproduction	pending
2	Dataset licensed and citable	pending
3	Training logs + final checkpoint committed	pending
4	Latency reproduced by the posted script (Pi)	pending
5	Held-out test set with unseen speakers	pending
6	Baseline of comparable size compared	pending

Honesty note: out-of-scope rejection above is measured only on the OOS takes inside this take bank; a quiet-room false-trigger test was not run. Accuracy figures describe these recordings, not unseen microphones.